Effect of Sleep Duration and Concussion History on Neurocognitive Testing Symptoms for Post-injury States

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Abstract

Objective: Evaluate the influence of sleep duration and concussion history on postconcussion symptoms in adolescent athletes. **Design:** Observational retrospective study using the Immediate Post-Concussion Assessment and Cognitive Testing and Post-Concussion Symptom Scale (PCSS). Multivariable linear regression assessed sleep hours against 22 PCSS symptoms, controlling for demographic and health variables. **Setting:** Urban concussion centers in Colorado and Florida, 2009 to 2019. **Participants:** 11 564 student-athletes aged 12 to 22, categorized by concussion history. **Interventions:** Analysis of sleep duration and concussion history in relation to neurocognitive and psychiatric symptom severity. **Main Outcome Measures:** Primary outcomes included neurocognitive, psychiatric, and total symptom scores. Secondary outcomes were specific PCSS symptoms. **Results:** Among 5349 student-athletes, 2671 (49.9%) had no prior concussions and 2678 (50.1%) had 1 or more. For those without prior concussions, sleep was negatively associated with age ($\beta = -0.18$, 95% confidence interval [CI], -0.22 to -0.13, $P < 0.0001$), vomiting ($\beta = -0.22$, 95% CI, -0.38 to -0.05, $P = 0.012$), and difficulty concentrating ($\beta = -0.11$, 95% CI, -0.19 to -0.35, $P = 0.005$). In athletes with a history of concussion, less sleep correlated with decreased age ($\beta = -0.11$, 95% CI, -0.14 to -0.07, $P < 0.0001$), headache ($\beta = -0.065$, 95% CI, -0.12 to -0.01, $P = 0.031$), irritability ($\beta = -0.08$, 95% CI, -0.15 to -0.01, $P = 0.021$), and difficulty concentrating ($\beta = -0.08$, 95% CI, -0.15 to -0.01, $P = 0.031$) but increased sensitivity to light ($\beta = 0.10$, 95% CI, 0.001–0.137, $P = 0.048$), numbness/tingling ($\beta = 0.15$, 95% CI, 0.04–0.26, $P = 0.008$), and feeling slowed down ($\beta = 0.13$, 95% CI, 0.05–0.21, $P = 0.001$). **Conclusion:** Sleep duration and concussion history are associated with variations in postconcussion symptom severity among adolescent athletes. This data underscore the need for individualized management strategies based on sleep patterns and concussion history.

Key Words: concussion, sleep duration, adolescent athletes, impact testing, post-concussion recovery

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INTRODUCTION

Adolescent participation in athletics is associated with improved physical, mental, and cognitive health outcomes.[1,2] These health benefits are evident in the substantial number of adolescents—approximately 8.8 million—actively participating in organized high school athletics across the United States.[3] In tandem, the prevalence of sport-related concussions has increased as contact-sport players face a 20% risk of such injury per year.[4] Of the 1.7 million concussions reported annually among adolescents by the Centers for Disease Control and Prevention, approximately 20% are attributed to sports participation.[5,6] Therefore, despite the reported health benefits of adolescent sports participation, the high incidence of concussions poses significant risks.

The impact of sports-related concussions on the developing brain of adolescents can be severe and long-lasting.[7] While most adolescents make a full recovery within 4 weeks, approximately 10% to 33% of patients experience persistent postconcussive symptoms beyond this period, which may include headache, nausea, vomiting, sensitivity to light, difficulty concentrating, and sleep difficulties.[8,9] With the rising incidence of concussions and their effects on adolescent development, the importance of sleep in the medical management of persistent postconcussive symptoms has garnered considerable attention. Sleep deficiency, which is generally negatively associated with neurocognitive and psychological recovery, has been shown to increase the severity and duration of postconcussion symptoms significantly.[10] However, there has been a paucity of data investigating how pre- and postconcussion sleep patterns are associated with presenting symptomatology in adolescents, particularly in those with a history of multiple concussions.

Given the potential influence of sleep on recovery from concussion symptoms, this study specifically examines the

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effects of sleep duration on postinjury neurocognitive testing in a large cohort of athletes. Focusing on the postconcussion phase, the study evaluates the effect of sleep duration on symptoms between athletes with and without a history of previous concussion.

METHODS

Study Population and Data Collection

This study retrospectively analyzed an Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT) and Post-Concussion Symptom Scale (PCSS) registry of 11 564 student-athletes who reported to multiple large urban concussion centers in Colorado and Florida between July 2009 and June 2019.[11] Eligibility for inclusion in the study was determined by age ≥12 and ≤22 years, middle school or high school enrollment, and ≥1 season of athletic participation. Patients with incomplete demographic data or any invalid ImPACT test were excluded from the study. This study specifically analyzed data from postconcussion assessments to understand the impact of sleep duration on recovery and symptom severity, excluding baseline preinjury data.

Demographics, clinical history, 7-point PCSS survey, and ImPACT test data were collected, as previously described.[12] Demographic data included age, sex, hours of sleep the night before evaluation, history of prior concussion diagnosis, history of diagnosed learning disability (DLD), and history of diagnosed attention-deficit/hyperactivity disorder (ADD/ADHD). Concussions were defined as direct blows to the head, face, or neck, resulting in rapid-onset neurological dysfunction and/or a range of clinical signs and symptoms that may or may not involve loss of consciousness.[13]

Concussion Symptom Assessment

The PCSS survey, previously validated in adolescent athlete populations,[14] was used to quantify pre- and postconcussion symptoms on a 7-point Likert scale from absent (0) to severe (6). This 22-symptom survey is categorized into migraine, cognitive, neuropsychiatric, and sleep clusters.[12] Individual symptom measures included headache, vomiting, nausea, balance problems, dizziness, fatigue, sensitivity to light, drowsiness, sensitivity to noise, and visual problems. Psychiatric Symptoms were grouped with the variables of irritability, nervousness, sadness, feeling more emotional, numbness or tingling, feeling mentally foggy, feeling slowed down, difficulty concentrating, and difficulty remembering.

Neurocognitive Testing

The ImPACT test, previously validated in adolescent athlete populations,[11] was administered to assess neurocognitive performance postinjury. Immediate Post-Concussion Assessment and Cognitive Testing reports 4 composite scores: verbal memory, visual memory, visual-motor processing speed, and reaction time. The test was used to provide baseline neurocognitive levels and/or evaluate changes from baseline testing in the study participants. ImPACT test was performed within the first 72 hours after head injury as per ImPACT protocol.

Statistical Analysis

The cohort was divided into 2 groups: participants without a history of prior concussions and those with a history of 1 or more concussions. Hours of sleep were divided into short (<7 hours of sleep), intermediate (7–9 hours), and long (>9 hours), based on previous studies.[15,16] Normally distributed continuous variables were presented as mean ± SD and examined with Student's t tests. Sleep duration was the primary dependent variable in statistical analyses. Variables significant on univariate analysis with $P < 0.05$ were advanced into multivariable analysis. Thus, in the multivariable analysis, ADHD, DLD, Age, Sex, and number of past concussions were all controlled for in the linear regression analysis. All statistical analyses, including eta squared for effect size calculation, were performed using GraphPad Prism version 10.0.3 (San Diego, CA).[17]

Ethics Statement

Given the retrospective, de-identified nature of the data, the need for consent for participation in the study was waived. Retrospective analysis of the data set was approved by the Institutional Review Board at the Icahn School of Medicine at Mount Sinai. The study was conducted according to the retrospective strengthening the reporting of observational studies in epidemiology reporting guidelines.[18]

RESULTS

The study cohort consisted of 5349 student-athletes, with 2671 (49.9%) having no history of prior concussions and 2678 (50.1%) with 1 or more prior concussions. There existed several demographic differences between the 2 groups, as reported in Table 1. In particular, student-athletes with a history of concussion were significantly more likely to be male (66.8% vs 59.3%; $P < 0.0001$), exhibit higher rates of DLD (4.0% vs 2.9%; $P < 0.0001$), and ADHD (7.5% vs 4.5%; $P < 0.0001$) compared with those without a history of concussion. On the night before ImPACT assessment, participants without a history of concussion had marginally more hours of sleep than those with a concussion history (7.7 ± 1.8 vs 7.6 ± 1.7; $P < 0.005$). In addition, those without a history of concussion exhibited higher symptom scores (21.5 ± 21.7 vs 15.1 ± 19.0; $P < 0.0001$), neurologic symptoms (11.3 ± 10.9 vs 7.9 ± 9.9; $P < 0.0001$), and psychiatric symptoms (7.7 ± 9.6 vs 5.4 ± 8.0; $P < 0.0001$) when compared with student-athletes with a history of concussion.

Hours of sleep were then placed into 3 categories—short (<7 hours of sleep), intermediate (7–9 hours), and long (>9 hours). Greater duration of sleep was associated with fewer neurologic and psychiatric symptoms postinjury in both those with a history of concussion and those without (Table 2). For all sleep duration categories, student-athletes without a prior history of concussion showed higher neurologic symptom scores (low sleep $\eta^2 = 0.028$, medium sleep $\eta^2 = 0.022$, and high sleep $\eta^2 = 0.030$), psychiatric symptom scores (low sleep $\eta^2 = 0.021$, medium sleep $\eta^2 = 0.015$, and high sleep $\eta^2 = 0.019$), and total symptoms (low sleep $\eta^2 = 0.012$, medium sleep $\eta^2 = 0.011$, and high sleep $\eta^2 = 0.003$) after their first concussion as compared with athletes with a history of concussion (Figure 1). This study defined η^2 of <0.01 as

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TABLE 1. Demographic and Clinical Characteristics

	No History (N = 2671) (SD)	History (N = 2678) (SD)	*P*	η^2
Age (mean, SD)	15.7 (2.0)	15.9 (1.9)	**<0.0001**	0.004
% Female	40.7 (0.5)	33.2 (0.47)	**<0.0001**	0.006
DLD	2.9 (0.2)	4.0 (0.2)	**0.0172**	0.001
ADD/ADHD	4.5 (0.2)	7.5 (0.3)	**<0.0001**	0.004
Hours of sleep	7.7 (1.8)	7.6 (1.7)	**0.0046**	0.002
Past concussions	0	1.5 (0.9)		
Total symptom score	21.5 (21.7)	15.1 (19.0)	**<0.0001**	**0.024**
Neurologic symptoms	11.3 (10.9)	7.9 (9.9)	**<0.0001**	**0.026**
Psychiatric symptoms	7.7 (9.6)	5.4 (8.0)	**<0.0001**	**0.018**

Bold values are statistically significant to $P < 0.05$.

a small effect size, <0.06 as a medium, and >0.14 as a large effect, so all symptom scores fall within the small to medium effect size range.[19]

A multivariable linear regression analysis was performed to determine whether neurologic and psychiatric symptoms were associated with hours of sleep in those student-athletes with and without a history of concussion (Table 3). For student-athletes with no prior concussion, shorter sleep trended with an increase in neurologic symptoms ($\beta = -0.0046$, 95% CI, −0.01704 to 0.007832, $P = 0.4678$) and psychiatric symptoms ($\beta = -0.004436$, 95% CI, −0.01879 to 0.009919, $P = 0.5446$), although these associations were not statistically significant. Nevertheless, the total symptom score did significantly increase with decreased sleep ($\beta = -0.007149$, 95% CI, −0.01120 to −0.003095, $P = 0.0006$), a pattern also observed in those with a history of concussion ($\beta = -0.008934$, 95% CI, −0.01243 to −0.005439, $P < 0.0001$).

Next, this study explored the specific symptoms associated with sleep duration for both groups of student-athletes using a multivariable linear regression analysis (Table 4). Student-athletes without a history of concussion with shorter sleep durations showed an association with vomiting ($\beta = -0.2157$, 95% CI, −0.3831 to −0.04816, $P = 0.0116$) and difficulty concentrating ($\beta = -0.1135$, 95% CI, −0.1919 to −0.03508, $P = 0.0046$), with a trend of improvement in these symptoms as sleep duration increased. Conversely, those with a history of concussion and shorter sleep durations experienced several symptoms that worsened with longer sleep: headache ($\beta = -0.06533$, 95% CI, −0.1247 to −0.005939, $P = 0.0311$), sensitivity to light ($\beta = 0.06870$, 95% CI, 0.0006526–0.1368, $P = 0.0478$), irritability ($\beta = -0.08084$, 95% CI, −0.1494 to −0.01229, $P = 0.0208$), numbness ($\beta = 0.1504$, 95% CI, 0.03909–0.2616, $P = 0.0081$), and feeling slowed down ($\beta = 0.1321$, 95% CI, 0.05147–0.2126, $P = 0.0013$), while difficulty concentrating improved ($\beta = -0.07881$, 95% CI, −0.1504 to −0.007260, $P = 0.0309$).

DISCUSSION

Amidst growing concerns about sport-related concussions in adolescent athletes and the critical role of sleep in their recovery, this study aimed to supplement the limited medical literature on how sleep patterns affect concussion symptomatology in young athletes by distinguishing between those with and without a history of concussion. The fundamental question of this study was whether sleep duration was associated with symptomatologic changes in student-athletes with and without a history of prior concussions.

Demographic Differences and Concussion History

In comparing the demographic profiles of student-athletes with a history of concussion to those without such a history, several differences were identified. The group encapsulating

TABLE 2. Symptom Scores Based on Hours Slept

	No History (N = 2671) (SD)	History (N = 2678) (SD)	*P*	η^2
Neurologic symptoms				
<7 h (N = 1406)	11.6 (10.6)	8.1 (10.0)	**<0.0001**	0.028
7–9 h (N = 3139)	11.1 (11.1)	7.9 (9.9)	**<0.0001**	0.022
>9 h (N = 803)	10.5 (10.6)	7.0 (9.3)	**<0.0001**	0.030
Psychiatric symptoms				
<7 h (N = 1406)	8.0 (9.2)	5.4 (7.8)	**<0.0001**	0.021
7–9 h (N = 3139)	7.5 (9.7)	5.3 (7.9)	**<0.0001**	0.015
>9 h (N = 803)	7.1 (8.9)	4.8 (7.5)	**0.0008**	0.019
Total symptom score				
<7 h (N = 1406)	25.1 (22.3)	20.2 (20.8)	**<0.0001**	0.012
7–9 h (N = 3139)	16.1 (18.6)	12.2 (17.2)	**<0.0001**	0.011
>9 h (N = 803)	21.4 (20.5)	18.9 (21.3)	0.1520	0.003

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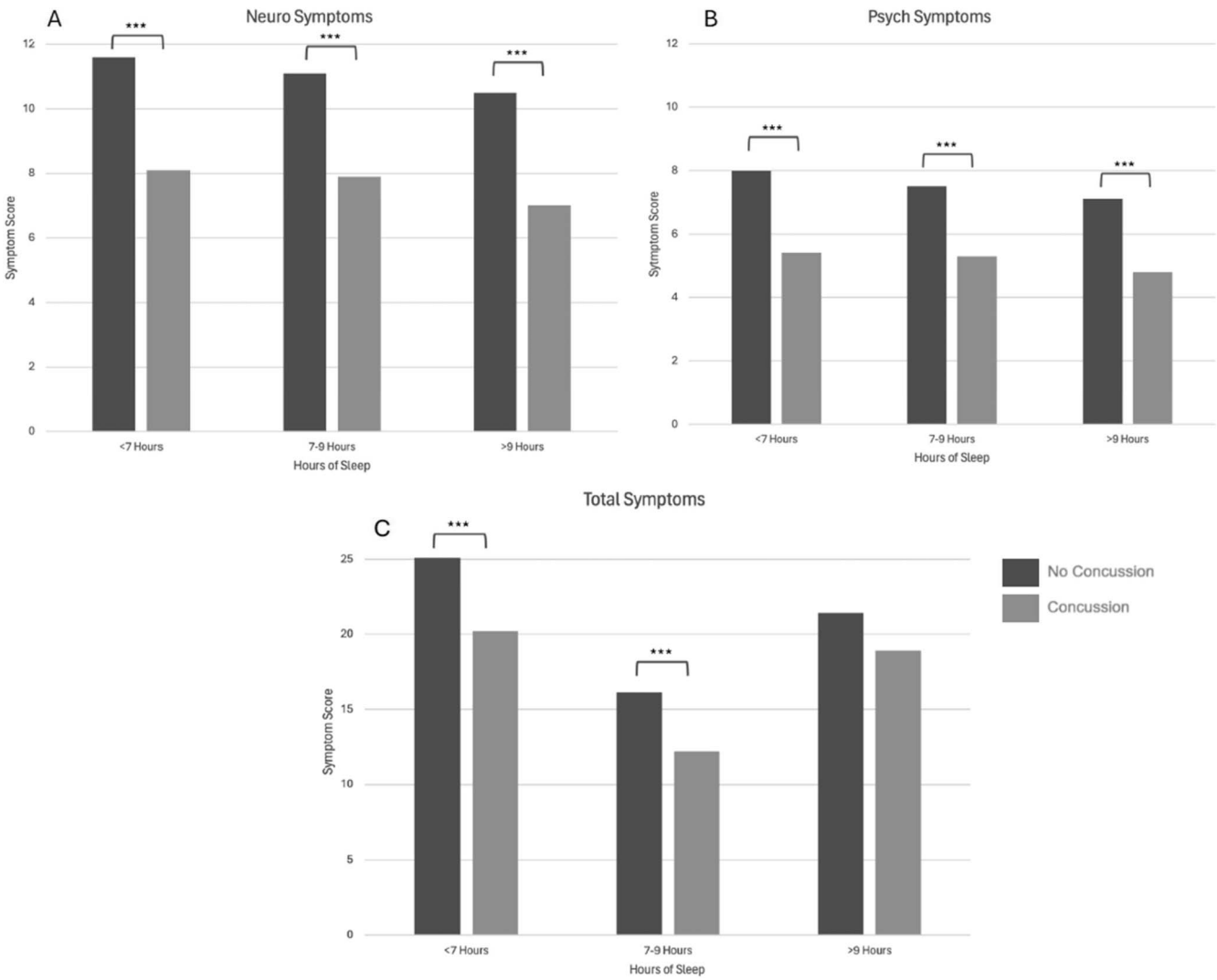

Figure 1. Comparison of mean symptom scores in relation to hours of sleep (<7 hours, 7–9 hours, >9 hours) in adolescent athletes: A, neurological symptoms, B, psychiatric symptoms, and C, total symptoms. Dark bars represent athletes with no concussion history, while light bars represent athletes with a history of concussions. In all panels, *** indicates a statistically significant difference ($P < 0.001$) between the groups compared.

athletes with a history of concussion had a higher percentage of male athletes, athletes with DLD, and athletes with ADD/ADHD. The higher percentage of male athletes with a history of concussion in this study could be due to the very nature of men's athletics as compared with women's athletics. Despite the higher-energy impacts often associated with men's sports, it is noteworthy that similar sports for females exhibit equal or higher concussion rates, even in those with approximately equal participation across sexes, such as soccer.[20] Furthermore, it is unclear why athletes with learning disabilities or ADD/ADHD are more likely to have had a concussion. Past studies have highlighted learning disorders and attention-deficit/hyperactivity disorders as risk factors for concussion recovery in young athletes,[21,22] yet this link between learning disabilities or ADD/ADHD and prolonged concussion recovery fails to fully explain the high incidence rate that appears in this study. It is worth noting that previous research presents study participants with ADHD as also having higher reported symptoms at baseline than those without this diagnosis.[23–25] A study by Kaye et al adds to this thinking by observing that the symptoms of ADHD closely resemble those that are often associated with concussions,[26] which may make the 2 difficult to distinguish from 1 another. Kaye et al hypothesize that there may exist a self-reporting bias in the population of study participants with ADHD as they may be

TABLE 3. Multivariable Linear Regression Neurologic, Psychiatric, and Total Symptom Score

	Estimate	95% CI	ltl	*P*
Neurologic symptoms				
No history	−0.004605	−0.01704 to 0.007832	0.7262	0.4678
History	−0.004708	−0.01622 to 0.006802	0.8021	0.4226
Psychiatric symptoms				
No history	−0.004436	−0.01879 to 0.009919	0.6061	0.5446
History	0.003128	−0.01131 to 0.01757	0.4247	0.6711
Total symptom score				
No history	−0.007149	−0.01120 to −0.003095	3.458	**0.0006**
History	−0.008934	−0.01243 to −0.005439	5.013	**<0.0001**

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TABLE 4. Multivariable Linear Regression Specific Symptoms

	Estimate	95% CI	ltl	*P*
Headache				
No history	−0.03153	−0.09918 to 0.03612	0.9141	0.3608
History	−0.06533	−0.1247 to −0.005939	2.157	**0.0311**
Vomiting				
No history	−0.2157	−0.3831 to −0.04816	2.525	**0.0116**
History	0.03251	−0.1023 to 0.1673	0.4729	0.6363
Sensitivity light				
No history	0.04747	−0.02867 to 0.1236	1.223	0.2216
History	0.06870	0.0006526 to 0.1368	1.980	**0.0478**
Irritability				
No history	−0.007721	−0.08330 to 0.06785	0.2004	0.8412
History	−0.08084	−0.1494 to −0.01229	2.313	**0.0208**
Numbness				
No history	−0.02334	−0.1405 to 0.09381	0.3908	0.6960
History	0.1504	0.03909 to 0.2616	2.650	**0.0081**
Slowed down				
No history	0.06264	−0.02471 to 0.1500	1.407	0.1597
History	0.1321	0.05147 to 0.2126	3.213	**0.0013**
Difficulty concentrating				
No history	−0.1135	−0.1919 to −0.03508	2.839	**0.0046**
History	−0.07881	−0.1504 to −0.007260	2.160	**0.0309**

more used to neuropsychological exams and reporting symptoms to medical professionals. We agree with their hypothesis, and this finding necessitates further targeted investigation to answer the question of why the data show a higher percentage of athletes with DLD or ADD/ADHD who have experienced prior concussions.

Differential Effect of Concussion on Symptom Severity

Interestingly, student-athletes without a history of concussion showed higher symptom scores, neurologic symptoms, and psychiatric symptoms as compared with those with a history of concussion. This finding contradicts similar studies that exist in the literature. For example, Covassin et al[27] studied the influence of concussion history, as defined by 2 or more concussions, and concussion recovery among collegiate athletes; the authors concluded that athletes with a history of head trauma take longer to recover several of their cognitive functions than athletes without such history. A similar study by Murdaugh et al[28] found that less sleep was correlated with increased cognitive and neuropsychological symptoms after sports-related concussions. A third study, conducted by Brown et al,[29] examined concussion-related symptoms among combat sports athletes with and without a history of concussion, finding that athletes with a history of concussion have a greater chance of presenting with higher symptom scores. Moran et al[30] demonstrated a higher total number of symptoms and increased symptom severity with a history of concussion among collegiate athletes. It is clear that the bulk of the literature points to this link between more severe or persistent symptoms and having a history of concussion.

However, there exist several studies where young athletes with a history of concussion perform better on components of neurological testing than their counterparts with no history of concussion, which more closely mirror our findings.[6,26] We hypothesize that this result could be due to a process of normalization of concussion symptoms or, as previously described,[6] compensation over time whereby those with a history of concussion have become inured to symptoms and are likely to underreport them in comparison to those who have never experienced a concussion. To expand upon this idea, we posit that athletes with concussion histories may have unresolved symptoms that tend to linger and are thus less likely to report much of a difference upon sustaining a new concussion. This notion of desensitization deserves future examination, which does not exist in the literature. Another possible reason for underreporting concussions could include simple yet powerful social forces such as team stigmatization or not wanting to miss competitions for the same injury—in this case, concussions. This social phenomenon among athletes is exemplified in a study by O'Connor et al[31] that examined, through surveys, concussions among jockeys and found that 2 of the principal reasons for continuing to ride despite having sustained a concussion were the desire not to let anyone down and considering not riding as a sign of weakness. To complement these data, there is evidence in the literature of widespread underreporting of concussion symptoms in athletes; the reasons for this could be vast and encompass a range of factors related to social perception, medical symptoms, or a combination of these 2 elements.[32] Regardless, this departure from an expected finding deserves further inquiry and discussion.

In examining the influence of sleep on the presentation of symptoms in concussed athletes, this study identified a link between improved neurologic and psychiatric symptoms and more sleep in both those with and without a history of concussion, although this link did not hold uniformly across all symptoms. Of note, the difference in sleep between student-athletes with and without a prior history of concussion was not clinically significant (7.7 vs 7.6 hours; $P < 0.005$). Past studies point to the importance of sleep in the recovery of concussions. For example, Hoffman et al[33] examine the connection between sleep and concussion recovery timelines in collegiate athletes, concluding that poorer sleep quality may be associated with a longer timeline of recovery from symptoms. Although this finding corroborates data presented in this study, the exact mechanism is not clear as to why sleep dysfunction may impair concussion recovery. This topic presents an avenue of potential future research.

Concussion Symptom Severity and Recovery Dynamics

The multivariate analyses in this study investigated a range of specific concussion symptoms associated with sleep for both those with and without a history of concussion, and we observed that difficulty concentrating negatively correlates with increasing sleep, indicating that sleep significantly improves concentration postconcussion. This same relationship can also be seen in several other symptom categories of this multivariate analysis, such as irritability and headache for those with a history of concussion and vomiting for those without a history of concussion. These findings fall in line with previous studies, as discussed previously, and display sleep as a powerful predictor of symptoms and symptom recovery,

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regardless of whether an athlete has experienced prior concussions. Chung et al[34] underscore and add to this point in their study of sleep and concussion recovery among adolescent athletes, concluding that poor sleep quality is strongly associated with greater symptom severity and longer recovery time. However, the importance of sleep as an indicator of recovery, or as a necessary component thereof, is slightly undermined by other aspects of this study. In particular, light sensitivity, numbness, and feeling slowed down were significantly associated with more hours of sleep for those with a history of concussion. One potential explanation for this finding is that concussion trauma induces neurological changes that specifically affect light sensitivity, numbness, and feeling slowed down, and these changes have yet to resolve or are somehow exacerbated by excessive sleep. The exact mechanisms to support this hypothesis have yet to be elucidated and merit further research. Another possibility for these findings concerns the timing of the concussion testing. For example, testing in proximity to a long sleep may increase sluggishness in respondents. It is thus important to note the details of the timing of concussion testing in the acquisition of future concussion data.

Existing literature on the topic of sleep disorders, as well as our study, indicate that these conditions are a common consequence of sports-related concussions,[35] yet this presents an obvious conundrum: while sleep may be a crucial component of concussion recovery, concussions can disrupt sleep patterns and stall recovery. Along with other concussion symptoms that may impair sleep patterns, such as nausea, dizziness, or headache, sleep disturbances stand to interfere with or delay recovery and present another complicating factor in the recovery process.[36] Mitigating the detrimental effects of concussion on sleep seems to be yet another way to improve concussion recovery and deserves further research attention.

Limitations and Future Directions

While maintaining the robustness of its sample size and findings, this study has several limitations. First, the retrospective methodology limits the findings' generalizability to the clinical setting and the causality of the relationship between sleep patterns and concussion symptoms. Second, the study's setting (Colorado and Florida) limits its generalizability to other geographical or demographic regions. However, these 2 geographically and climatically distinct states allow this study to capture variations in sports popularity and playing conditions. Third, self-reported sleep duration may introduce reporting bias, as with any self-assessment. In addition, while this study adjusted for several confounding variables available from the ImPACT postinjury database, other factors influencing concussion symptoms and recovery, such as the severity of the concussion, were difficult to account for. Future research could benefit from integrating measures of concussion severity to understand its impact on symptomatology and recovery better.

The study's data demonstration of lower symptom burden in student-athletes with a history of concussion, which appears counterintuitive from a clinical standpoint, aligns with a few publications leveraging ImPACT databases for this population. These findings suggest a complex relationship between concussion history and symptomatology that merits further exploration. In a similar manner, the results demonstrating that increased sleep correlated with increased symptom severity for a subset of symptoms may have been affected by the large size of the dataset and small effect sizes, highlighting areas of future investigation rather than diminishing the study's validity. Additional future research directions include randomized controlled trials examining the efficacy of sleep quality and quantity in concussed athletes, which could provide evidence for clinical practice recommendations as well as the long-term effects of concussion on sleep patterns and neurologic and psychiatric functions.

CONCLUSIONS

Evaluating student-athletes after concussions remains a complex topic of investigation that demands the assessment of a multitude of risk factors. In this vein, this study highlights the association of increased sleep with a reduction in symptom severity, irrespective of concussion history, while also demonstrating that student-athletes may report higher symptom scores after their first concussion. Furthermore, the differential impact of concussion history on symptomatology after a new concussion emphasizes the need for more personalized management plans for student-athletes. Therefore, the data in this study suggest that these 2 factors—sleep and concussion history—should be considered when assessing concussion symptoms, performing baseline testing, and designing treatment plans.

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